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BAGONG PILIPINAS

CC 101 – Introduction to Computing

Laboratory Activity No. 1 | Week 2: Introduction to Computing (CC1O1) | Unit 2, Computing Career Pathways

Explore Career Pathways

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Course/Program: BSCS 1-1 **Date:** 09/16/26

I. Objectives

At the end of this activity, you should be able to:

- Identify the major computing career pathways introduced in lecture and describe what each one involves.
- Choose one computing career pathway that best matches your interests and explain why.
- Map a realistic entry-to-expert roadmap for your chosen pathway, including representative job titles and key skills at each stage.
- Compare your current skills against your chosen pathway's requirements and identify a concrete plan to close the gap.

II. Materials Needed

- Your CC 101 Unit 2 lecture slides/notes on Computing Career Pathways
- This worksheet (printed copy, or the digital file to fill out and submit)
- Optional: internet access, for Part E (real job posting search)

III. Background

Computing is not just "programming." As introduced in the Unit 2 lecture, the field branches into several distinct career pathways, each with its own operational scope, required technical competencies, representative job titles, and professional standard or reference body: Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, Cybersecurity & Threat Defense, and Information Systems & Enterprise IT. Each pathway also has its own career roadmap — the entry-level, mid-level, senior-level, and expert/leadership roles a professional typically grows through. This activity asks you to review those five pathways, choose the one that interests you most, and map out what growing in that pathway could look like for you.

IV. Procedure

Part A – Recall the Five Pathways

1. Using your lecture notes or slides, complete Table 1 by writing the core operational scope and one to two representative job titles for each of the five computing career pathways.

Part B – Choose Your Pathway

2. From the five pathways, choose ONE that interests you the most.
3. In two to three sentences, explain why you chose it (e.g., skills you already enjoy, a subject you like, a job title that excites you).

Part C – Map Your Career Roadmap

4. If your chosen pathway is Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, or Cybersecurity & Threat Defense, use the career roadmap shown for it in the lecture. If

you chose Information Systems & Enterprise IT (no roadmap was shown in lecture), research typical entry-level, mid-level, senior-level, and expert/leadership roles for this pathway online.

5. Complete Table 2 with example job titles and key skills/responsibilities at each of the four career stages.

Part D – Skill Gap Self-Assessment

1. Looking at the essential technical competencies for your chosen pathway (Table 1 / lecture), list at least five in Table 3. For each, mark whether you already have some experience with it, and briefly note how you plan to learn it if you don't.

Part E – Real-World Job Search (Optional / Bonus)

2. Search a real job platform (e.g., LinkedIn, Indeed, JobStreet, Kalibrr) for one entry-level job posting that matches your chosen pathway. Record it in Table 4.

V. Data and Findings

Table 1. The Five Computing Career Pathways (Recap — Part A)

Pathway	Core Operational Scope	Representative Job Title(s)
Software Engineering	Designing, developing, testing, and maintaining software applications, frameworks, and systems architectures.	• Full-Stack Developer • Mobile App Developer • QA Automation Engineer
Systems & Cloud Infrastructure	Managing, configuring, and scaling physical/virtual servers, cloud platforms, network networks, and deployment pipelines.	• Cloud Architect • DevOps Engineer • Systems Administrator
Data Science & AI Engineering	Building predictive models, training machine learning algorithms, and processing large datasets to derive business insights.	• Data Scientist • Machine Learning Engineer • Data Architect
Cybersecurity & Threat Defense	Protecting digital assets, analyzing vulnerabilities, monitoring network threats, and implementing defensive security protocols.	• Security Analyst • Penetration Tester (Ethical Hacker) • Incident Responder
Information Systems & Enterprise IT	Aligning business strategies with technical tools, managing enterprise software deployments, and handling day-to-day tech support.	• IT Business Analyst • IT Project Manager • Systems Analyst

My Chosen Pathway (Part B)

I am most interested in:

Why I chose it:

Table 2. My Career Roadmap for My Chosen Pathway (Part C)

Career Stage	Example Job Title(s)	Key Skills / Responsibilities
Entry-Level	Associate, Coordinator, Junior Developer, Assistant	Learning core technical tools, executing daily tasks, supporting team projects, time management
Mid-Level	Specialist, Engineer, Consultant, Senior Analyst	Managing independent projects, data analysis, problem-solving, mentoring junior staff
Senior-Level	Manager, Lead, Principal, Senior Consultant	Strategic planning, team leadership, cross-department collaboration, budget management
Expert / Leadership	Director, VP, Chief Officer (CXO), Principal Expert	Setting company vision, high-stakes decision making, executive leadership, industry thought leadership

Table 3. Skill Gap Self-Assessment (Part D)

Essential Competency	Already Have It? (Y/N)	How I Plan to Learn It
Mobile App Development	N	Watch YouTube tutorials on Flutter or React Native and build a basic mobile app.
Data Science	N	Learn Python library basics like Pandas and NumPy through free Kaggle courses.
Incident Response	N	Study common cyberattack case studies and review basic security logging concepts online.
DevOps Engineering	N	Watch introductory videos on Git and Docker, then use them in a school project.
QA Automation Engineering	N	Take a free online course on Selenium and practice writing basic test scripts.

Table 4. Real Job Posting Found (Part E, Optional / Bonus)

Item	Details
Job Title	Junior Associate Software Engineer

Item	Details
Company	Accenture Philippines
Platform / Source (e.g., LinkedIn, Indeed, JobStreet)	Linkdln/ Jobstreet
Required Skills Listed (3–5)	1. Basic programming knowledge (Java, C++, C#, or .NET) 2.Foundational understanding of Software Development Life Cycle (SDLC) & Agile 3. Problem-solving and analytical thinking 4. Good communication and collaboration skills 5. Willingness to be trained in new coding languages and cloud tools

VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Why did you choose this particular pathway over the other four?

My reason for choosing software engineering is that I enjoy learning about programming and it's also because of my interest in technology. And this path feels like the best fit for my personal goals.

2. Looking at your roadmap, which career stage excites you the most, and why?

It excite me to became a junior associate because, it will mark the start of my career journey. I'm looking forward to be trained and learn new things in new coding language and work in modern cloud tools.

3. Which required skill or competency for this pathway do you already have, and which one will be hardest for you to develop?

The required skill I already have is "Willingness to be trained in new coding languages and cloud tools" and "Basic Programming knowledge" . But the one that hardest for me to develop is the "Good communication and collaboration skills" since i easily get nervous when talking to someone.

4. How does mapping this roadmap change — or confirm — your plans for the rest of this course and your degree?

Mapping this roadmap confirms that i choose the right degree for may career choice. And with that all I need to do is focus on a specific course to improve my skills.

VII. Scoring Guide

Criteria	Description	Points
Completeness	Tables 1–3 and the reflection are fully filled out (Table 4 is bonus).	25
Roadmap Accuracy & Depth	Roadmap and skills recorded match the lecture pathway (or credible outside research for IS & Enterprise IT).	25
Realization / Reflection	Answers are thoughtful, specific to the student's own choice, and well written.	30
Neatness & Timeliness	Worksheet is legible/well-formatted and submitted on time.	20
Total		**100**

Checked by: **Score:** / 100

Instructor's Signature over Printed Name / Date